

All computer viruses cause direct or indirect economic damages. Based on this, there are two categories of viruses:

- 1) Those that only disseminate and don't cause intentional damage
- 2) Those which are programmed to cause damage.

However, even by disseminating, they take up plenty of memory space, and time and resources that are spent on the clean-up job. Direct economic damages are caused when viruses alter the information during digital transmission. Considerable expenses are incurred by individuals, firms and authorities for developing and implementing the anti-virus tools to protect computer systems.

3. Logic bombs

A logic bomb, also known as “slag code”, is a malicious piece of code which is intentionally inserted into software to execute a malicious task when triggered by a specific event. It's not a virus, although it usually behaves in a similar manner. It is stealthily inserted into the program where it lies dormant until specified conditions are met. Malicious software such as viruses and worms often contain logic bombs which are triggered at a specific payload or at a predefined time. The payload of a logic bomb is unknown to the user of the software, and the task that it executes unwanted. Program codes that are scheduled to execute at a particular time are known as “time-bombs”. For example, the infamous “Friday the 13th” virus which attacked the host systems only on specific dates; it “exploded” (duplicated itself) every Friday that happened to be the thirteenth of a month, thus causing system slowdowns.

Logic bombs are usually employed by disgruntled employees working in the IT sector. You may have heard of “disgruntled employee syndrome” wherein angry employees who've been fired use logic bombs to delete the databases of their employers, stultify the network for a while or even do insider trading. Triggers associated with the execution of logic bombs can be a specific date and time, a missing entry from a database or not putting in a command at the usual time, meaning the person doesn't work there anymore. Most logic bombs stay only in the network they were employed in. So in most cases, they're an insider job. This makes them easier to design and execute than a virus. It doesn't need to replicate; which is a more complex job. To keep your network protected from the logic bombs, you need constant monitoring of the data and efficient anti-virus software on each of the computers in the network.